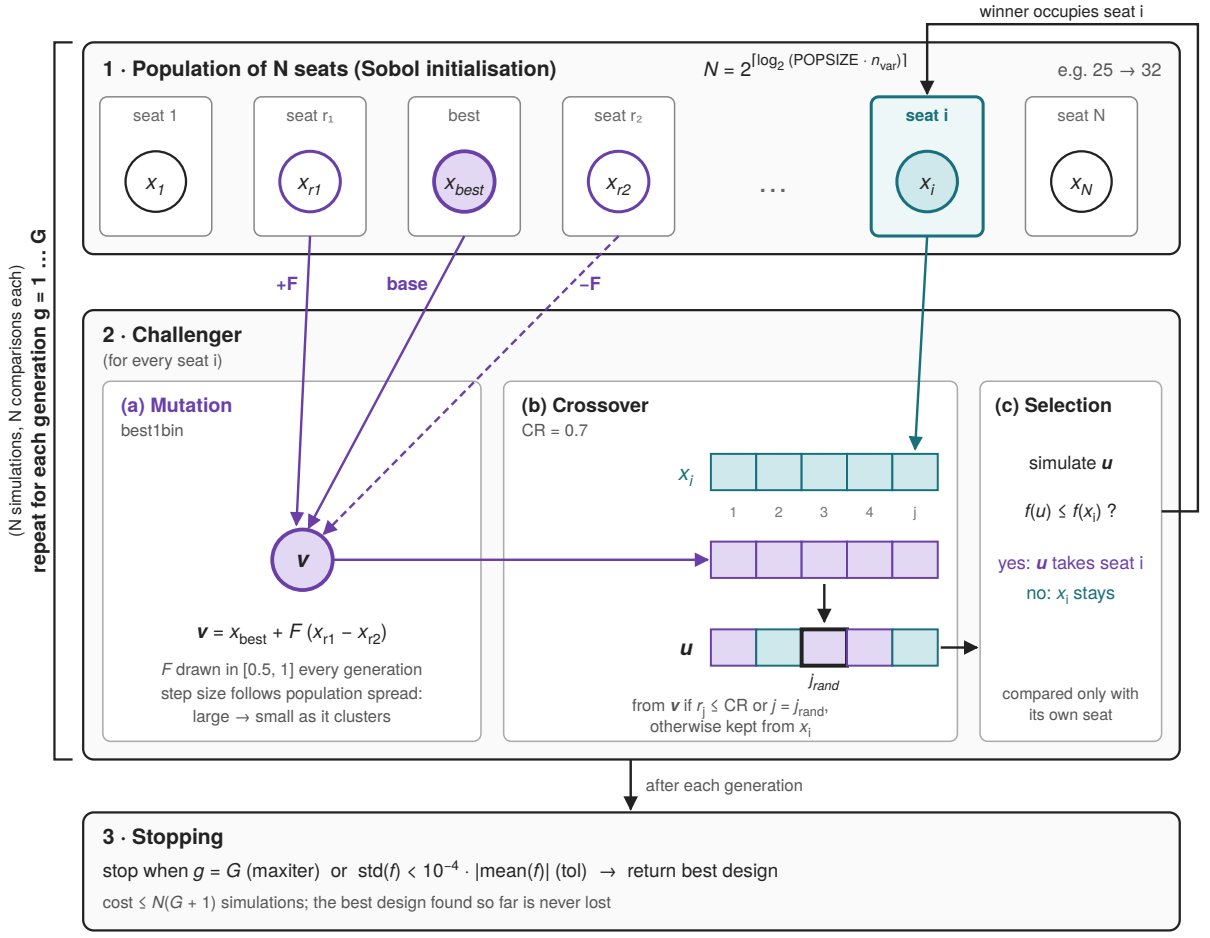
winner occupies seat i
1 · Population of N seats (Sobol initialisation)
$N = 2^{\lceil \log_2 (\text{POPSIZE} \cdot n_{\text{var}}) \rceil}$
e.g. 25 → 32
seat 1
seat r_1
best
seat r_2
seat i
seat N
x_1
x_{r1}
x_{best}
x_{r2}
x_i
x_N
+F
base
–F
2 · Challenger
(for every seat i)
(a) Mutation
best1bin
v
$\boldsymbol{v} = x_{\text{best}} + F\,(x_{r1} - x_{r2})$
F drawn in [0.5, 1] every generation
step size follows population spread:
large → small as it clusters
(b) Crossover
CR = 0.7
x_i
1 2 3 4 j
u
j_{rand}
from v if $r_j \le$ CR or $j = j_{\text{rand}}$,
otherwise kept from x_i
(c) Selection
simulate u
$f(u) \le f(x_i)$?
yes: u takes seat i
no: x_i stays
compared only with
its own seat
(N simulations, N comparisons each)
repeat for each generation g = 1 … G
after each generation
3 · Stopping
stop when $g = G$ (maxiter) or $\text{std}(f) < 10^{-4} \cdot |\text{mean}(f)|$ (tol) → return best design
cost ≤ $N(G + 1)$ simulations; the best design found so far is never lost